## Practical 1

## Aim: WAP to swap values of two variable using function and pointer.

**Input:**

```
#include <iostream>
using namespace std;

// Function to swap two numbers using pointers
void swapValues(int *x, int *y) {
   int temp;
   temp = *x;   // store value of x in temp
   *x = *y;     // assign value of y to x
   *y = temp;   // assign value of temp (original x) to y
}

int main() {
   int a, b;

   cout << "Enter value of a: ";
   cin >> a;
   cout << "Enter value of b: ";
   cin >> b;

   cout << "\nBefore swapping:" << endl;
   cout << "a = " << a << ", b = " << b << endl;

   // calling the swap function
   swapValues(&a, &b);

   cout << "\nAfter swapping:" << endl;
   cout << "a = " << a << ", b = " << b << endl;

   return 0;
}
```

**Output:**

```
Enter value of a: 32
Enter value of b: 35

Before swapping:
a = 32, b = 35

After swapping:
a = 35, b = 32
```

## Practical 2

## Aim: WAP to find length of string (UDF and Pointer).

**Input:**

```
#include <iostream>
using namespace std;

// User Defined Function to find length of string using pointer
int stringLength(char *str) {
   int length = 0;

   // loop runs until null character '\0' is found
   while (*str != '\0') {
      length++;
      str++;   // move pointer to next character
   }

   return length;
}

int main() {
   char str[100];

   cout << "Enter a string: ";
   cin.getline(str, 100);  // read string including spaces

   int len = stringLength(str);

   cout << "Length of the string is: " << len << endl;

   return 0;
}
```

**Output:**

```
Enter a String: Padmaj
Length of string is: 6
```